

Enabling Instance Segmentation: A Semi-Automatic Method for Thermal Event Annotation

Bartłomiej Jabłoński^{1b}, Dariusz Makowski^{1b}, *Senior Member, IEEE*, Aleix Puig Sitjes^{2b},
Marcin Jakubowski^{1b}, and the Wendelstein 7-X Team

Abstract—Contemporary infrared imaging systems in thermonuclear fusion devices are preventing thermal overloads on plasma-facing components (PFCs) relying on the surface temperature. Automatic delineation and classification of thermal events would facilitate scene understanding, contributing to advanced machine protection, control, and physics exploration applications. However, the absence of image annotations, which require a significant amount of expert labor and are prone to inconsistencies, limits the use of deep learning computer vision methods in fusion devices. A semi-automatic annotation method based on deterministic infrared image processing is proposed to reduce annotation efforts while maintaining consistency. The method exploits discharge sequence properties to minimize expert involvement. It was evaluated on infrared images from the Wendelstein 7-X (W7-X) stellarator by comparing the generated annotation with manually prepared ground-truth annotations. The generated annotations have a high mean similarity to the manual annotations, measured with Sørensen–Dice coefficient (SDC), equal to 0.825 with a sample standard deviation of 0.030. Furthermore, a customized metric temperature over limit weighted SDC (tlwSDC), which weighs pixel severity based on the surface temperature relative to the PFC temperature limit, is proposed, and this mean similarity is equal to 0.904 with a sample standard deviation of 0.018. Encouraging results for an infrared image from the W Environment in Steady-state Tokamak (WEST) tokamak indicate that the method might be cross-device viable. The proposed semi-automatic method enabled the generation of an annotated image dataset and, consequently, the training of the first W7-X instance segmentation model.

Index Terms—Infrared image processing, instance segmentation, semi-automatic annotation, thermal event.

I. INTRODUCTION

TWO concepts for magnetic confinement of the hot burning plasma are thought to be the fastest route to thermonuclear fusion: tokamaks and stellarators. Both types of

reactor use a magnetic field to confine the plasma and keep it away from the first wall. However, controlled heat and particle removal require some plasma-wall interaction, typically using a divertor, a part of the first wall built to accommodate the high power and particle loads expected in the reactor. The rest of the plasma vessel is expected to receive much lower heat fluxes. In the case of Wendelstein 7-X (W7-X), a most advanced HELIAS-type stellarator in Greifswald, Germany, a so-called island divertor [1] utilizes a set of magnetic islands at the edge, which are intersected by the divertor target plates. This leads to the formation of a so-called strike-line (SL) on each of the divertors, where most of the heat and particles coming from the plasma are deposited. In facilities such as W7-X or ITER, imaging systems are essential to protect all plasma-facing components (PFCs), such as the divertor, in a plasma vessel. This is to prevent overloading or damage to any of the PFCs. An intelligent understanding of heat load patterns, i.e., thermal events in acquired infrared images, might contribute to the following applications and corresponding examples.

- 1) *Machine Protection*: discriminate between true overloads and nondamaging thermal events, e.g., reflections, to prevent premature discharge termination by real-time protection systems [2].
- 2) *Machine Control*: segment a SL approaching a divertor boundary to reposition it with control coils [3].
- 3) *Physics Exploitation*: identify discharges with thermal events of requested characteristics for further studies [4].

Deep learning has been proven to deliver state-of-the-art performance in computer vision tasks such as instance segmentation, which involves identifying and delineating individual objects [5]. However, it requires a large amount of annotated image data [6], where annotation of a single daily life scene might require over an hour on average [7]. In the context of infrared images from thermonuclear fusion experiments, the annotation process is prone to producing inconsistent annotations as sequences are complex, have a high framerate (100 Hz), and thermal events have fuzzy contours. Moreover, it requires expertise to interpret them, similar to medical image analysis [8]. Thus, it is impractical for crowd-sourcing annotation. Synthetic diagnostics for infrared thermography have not yet reached a state to produce complex thermal event scenes that could be incorporated into training [9]. For instance, it was demonstrated for filament detection, which was trained on synthetic visible spectrum images [10].

The issues listed above limit the applicability of supervised computer vision methods to thermonuclear fusion devices.

Manuscript received 13 September 2023; revised 9 February 2024; accepted 25 March 2024. Date of publication 26 April 2024; date of current version 9 December 2024. This work has been carried out within the framework of the EUROfusion Consortium, funded by the European Union via the Euratom Research and Training Programme (Grant Agreement No 101052200—EUROfusion), and in part by the Engineering Grant EEG21-17, and in part by the Polish Ministry of Science and Higher Education within the international co-financed project Projekt Międzynarodowe Współfinansowane [International Co-financed Projects (PMW)]. The review of this article was arranged by Senior Editor R. Chapman. (*Corresponding author: Bartłomiej Jabłoński.*)

Bartłomiej Jabłoński and Dariusz Makowski are with the Department of Microelectronics and Computer Science, Lodz University of Technology, 90-924 Łódź, Poland (e-mail: jablonskiba@dmcs.pl).

Aleix Puig Sitjes and Marcin Jakubowski are with the Max Planck Institute for Plasma Physics, 17491 Greifswald, Germany.

Color versions of one or more figures in this article are available at <https://doi.org/10.1109/TPS.2024.3382844>.

Digital Object Identifier 10.1109/TPS.2024.3382844

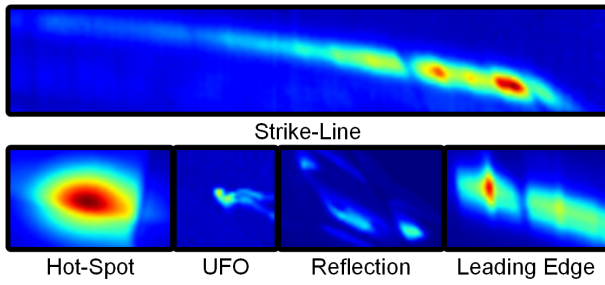


Fig. 1. Five examples of high-level thermal event categories in W7-X.

The only reported device to have an object detection (bounding boxes) database of real annotated thermal events is the W Environment in Steady-state Tokamak (WEST) tokamak [11].

To address the annotation bottleneck, weakly and semi-supervised approaches are active research topics [12]. These methods rely on box-level or image-level labels for training, which are easier to prepare than segmentation masks. However, they still require an existing database, and the resulting instance segmentation performance is reduced compared to fully supervised methods. As a consequence, semi-automatic methods designed for specific tasks are investigated [13].

The article presents a semi-automatic approach for infrared image annotation for thermonuclear fusion devices to accelerate the annotation process and impose consistency while integrating expert knowledge. The proposed hypothesis to verify is that a deterministic image processing method based on a reference image and max-tree representation significantly accelerates the thermal event annotation process by consistently generating annotations of a high degree of similarity to manual annotations in infrared images from nuclear fusion devices.

II. METHODS

A. Semi-Automatic Annotation

The method was primarily researched and developed on the 16-bit 1024×768 and 1280×1024 infrared images from the W7-X stellarator targeting thermal events illustrated in Fig. 1. The additional processing branch required for leading edge segmentation in a SL is not within the scope of this article. The annotation process has two stages: segmentation, i.e., pixel-precise delineation of potential thermal event instances, and classification, i.e., assignment of thermal event categories to the segmented instances.

The segmentation stage, often the most labor-intensive process in manual annotation preparation for a human, is automated by exploiting the property that thermal events are characterized by higher digital values, i.e., the surface temperature, in relation to their surroundings. A pixel value determines its relevance in contrast to color images, where a color value does not directly determine pixel relevance, and other features such as texture and shape are necessary. Therefore, pixel value analysis is sufficient to segment potential thermal event instances.

The classification stage, typically harder for an automated process than for a human, is delegated to an expert.

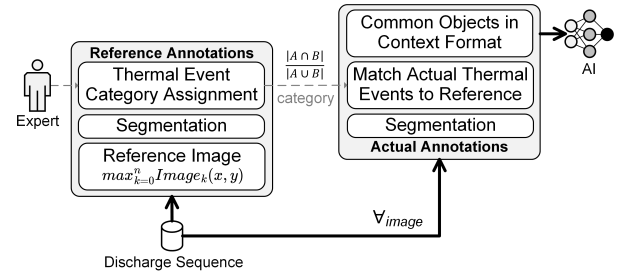


Fig. 2. Semi-automatic infrared sequence annotation pipeline. An actual segmented instance (A) is matched with a reference segmented instance (B) to copy its category.

However, to minimize expert effort and prevent classification inconsistencies between images within a discharge sequence, the proposed method exploits the property that current discharges are relatively static, the movement of thermal events is limited, and they appear and disappear in close proximity. Even more dynamic thermal events, for instance, caused by the injection of manipulators, Langmuir probes, or impurity powder droppers, have a fixed location, except for rare unforeseen unidentified flying objects (UFOs). Consequently, it was observed that the element-wise maximum of image pixels across all images in a discharge sequence produces a single coherent artificial infrared image. This image encapsulates the discharge progression by aggregating all the thermal events of the discharge. In the remainder of the article, this image is referred to as a reference image, and instances segmented in it are referred to as reference instances.

The proposed method is semi-automatic since it requires a human-in-the-loop (expert), as depicted in Fig. 2. The process starts with the reference image computation and its segmentation. The expert then assigns thermal event categories to the segmented reference instances. In addition, it is possible to mark a segmented reference instance as ignored if it is irrelevant, e.g., a visual artifact at the field of view (FoV) edge. Therefore, the segmentation process might produce excessive instances as long as they are not overwhelming for the annotator or interfere with other instances. Once the expert assigns categories to the segmented reference instances, actual discharge images are automatically segmented. Each actual segmented instance is automatically classified by comparing its overlap with the reference instances so that a category is copied from the matching reference instance with the most similar location, e.g., measured by intersection over union (IoU). Actual instances that do not overlap with any reference instance are discarded, e.g., spurious noise. Finally, the resulting annotations are converted and stored in Microsoft Common Objects in Context (MS COCO) format [14], which is suitable for instance segmentation. No parameters are exposed for modification to the expert so that all images and discharge sequences are uniformly processed, i.e., segmentation consistency is imposed.

B. Segmentation

The semi-automatic annotation method described in the previous paragraph can work with different segmentation procedures, whose performance determines the quality of

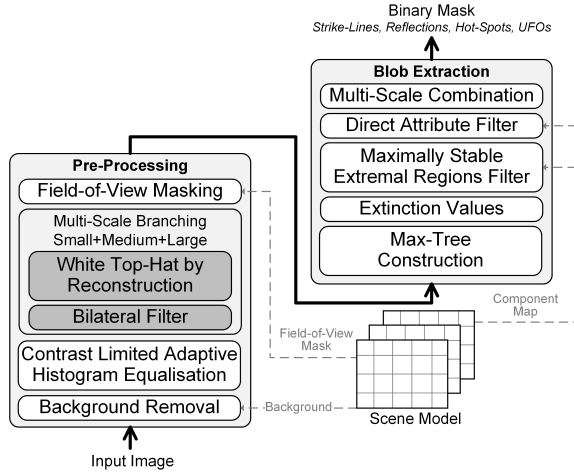


Fig. 3. Automatic infrared image segmentation pipeline. The scene model contains pixel-wise mappings of each FoV to the CAD model of the in-vessel components [15].

the final annotations. The article proposes a deterministic image-processing segmentation algorithm shown in Fig. 3.

1) *Pre-Processing*: A corrected background image for the background removal step is created from a measured background captured before a discharge and a minimum background that is a minimum of a temporal moving average window across a discharge sequence. Before operational phase (OP) 2.1 in W7-X, components were not actively cooled; therefore, residual heat from the previous discharge was observed in the measured background for the next discharge. If it is applied uncorrected, especially for a lower power discharge, some heat loads are canceled. The corrected background is reconstructed by erosion [16], where the measured background is a marker, and a pixel-wise minimum between the measured and minimum backgrounds is a mask.

Contrast-Limited Adaptive Histogram Equalization (CLAHE) improves local contrast by dividing the image into a grid and equalizing the histogram of each grid tile while limiting excessive amplification by clipping the histograms. The clip point is determined per image with a modified content-adaptive formula proposed in [17]. The modification involves computing a single clip point for the entire image rather than individually per tile and scaling the sum of weighted dynamic range and entropy terms by the number of grid tiles rather than the ratio of pixels in a block to its dynamic range. As a result of the automatic clip point adjustment, noise amplification is limited, particularly in the first frames of discharge sequences, where images have a low signal-to-noise ratio as components are still cold.

Edge-preserving smoothing bilateral filter [18] and multi-scale white top-hat by reconstruction [16] are applied to extract image details at different scales in parallel, i.e., using different filter and top-hat structuring element sizes [19]. The FoV mask is narrowed with morphological erosion to reduce the number of visual artifacts at the FoV edge.

2) *Blob Extraction*: The blob extraction is based on a max-tree representation [20], which was demonstrated to provide high performance for the SL segmentation [21], [22].

However, the proposed method extends it to other thermal events by introducing additional algorithms for this tree data structure. The extinction values [23] are computed from the height attribute. The max-tree leaves with height extinction values, which are above the minimum absolute threshold and the relative threshold with respect to the maximum extinction, are candidates for further processing with maximally stable extremal regions (MSER) [24], [25]. MSER determines, for each leaf, the connected component cutoff level for which the area attribute is the most stable.

Large cold uniform regions might be stable according to MSER. However, they cover smaller, more important, hotter regions or spread between PFCs. These regions are removed with the max-tree direct filtering [20] by the custom attribute computed for each max-tree node

$$f(v_p, v, c_p, c) = \begin{cases} 0, & \text{if } \frac{\min(v_p, v)}{\max(v_p, v)} < \frac{1}{3} \cup c_p \neq c \\ \max(v_p, v), & \text{otherwise} \end{cases} \quad (1)$$

where v is a pixel value, c denotes whether a pixel belongs to a divertor or any other PFC, and subscript p indicates the corresponding values for a parent node. Eventually, binary masks branching from multiscale white top-hat by reconstruction are combined to produce the final binary segmentation mask.

C. Evaluation

An evaluation dataset was prepared by manually annotating images¹ from 21 OP 1.2 discharge sequences covering nine immersion tube ports and different magnetic configurations to evaluate the quality of semi-automatic annotations. A single image acquired when the heating was stopped was selected from each discharge sequence. Manual annotations were created interactively by applying ClickSEG [26], which computes a segmentation mask based on manually set positive (including the object) and negative (not including the object) points within a manually drawn bounding box. Sorensen–Dice coefficient (SDC) [27], defined for two binary vectors (a, b)

$$\text{SDC}(a, b) = \frac{2 \sum_{i=0}^N a_i b_i}{\sum_{i=0}^N a_i + \sum_{i=0}^N b_i} \quad (2)$$

measures the similarity between the manual and generated annotations. In addition, a modified metric weighted by a real-valued vector (w)

$$w_i = \frac{T_i - T_i^{\text{Background}}}{T_i^{\text{Limit}}} \quad (3)$$

is proposed to compute temperature over limit weighted SDC (tlwSDC)

$$\text{tlwSDC}(a, b, w) = \frac{2 \sum_{i=0}^N a_i b_i w_i}{\sum_{i=0}^N a_i w_i + \sum_{i=0}^N b_i w_i}. \quad (4)$$

The proposed metric signifies the importance of higher temperatures relative to the corresponding PFC limit, reducing the impact of fuzzy colder edges.

¹Ground truth manual annotations are available at https://gitlab2.dmcs.pl/jablonskiba/w7x_ground_truth_manual_annotations.

TABLE I
SAMPLE MEAN AND STANDARD DEVIATION FOR SIMILARITIES BETWEEN GENERATED AND GROUND-TRUTH ANNOTATIONS

	Otsu's thresholding (baseline) [28]				Adaptive Gaussian thresholding [22]				Proposed method			
Mask	SDC	sSDC	tlwSDC	stlwSDC	SDC	sSDC	tlwSDC	stlwSDC	SDC	sSDC	tlwSDC	stlwSDC
Any	0.586	0.173	0.777	0.085	0.780	0.050	0.881	0.035	0.825	0.030	0.904	0.018
Strike-line	0.721	0.230	0.860	0.198	0.828	0.068	0.917	0.033	0.868	0.048	0.935	0.024
Reflection	0.108	0.126	0.194	0.189	0.679	0.072	0.761	0.073	0.748	0.044	0.818	0.044
Hot-spot	0.433	0.316	0.564	0.346	0.725	0.119	0.829	0.094	0.787	0.097	0.877	0.069

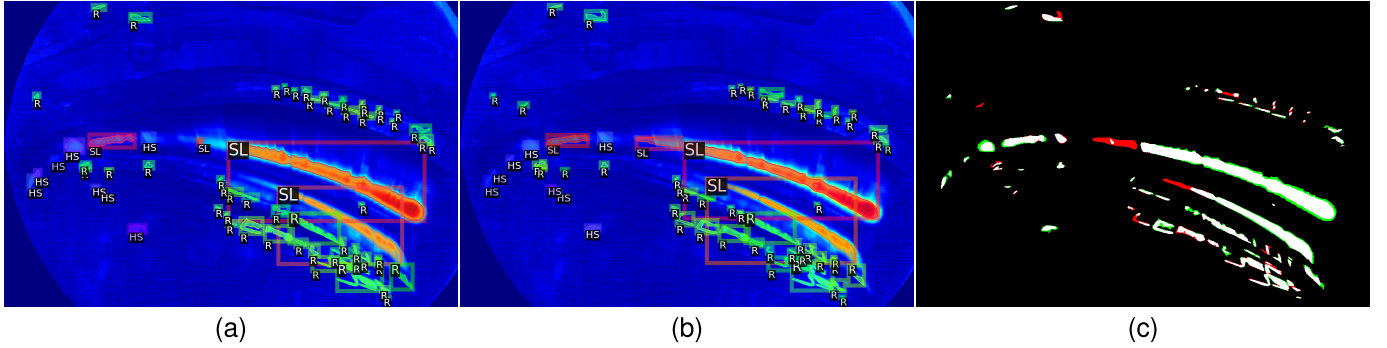


Fig. 4. 20181017.038 (AEF10), frame 960. (a) Generated annotations. (b) Ground truth annotations. (c) Difference (Sørensen–Dice coefficient (SDC): 0.805, temperature over limit weighted SDC (tlwSDC): 0.896).

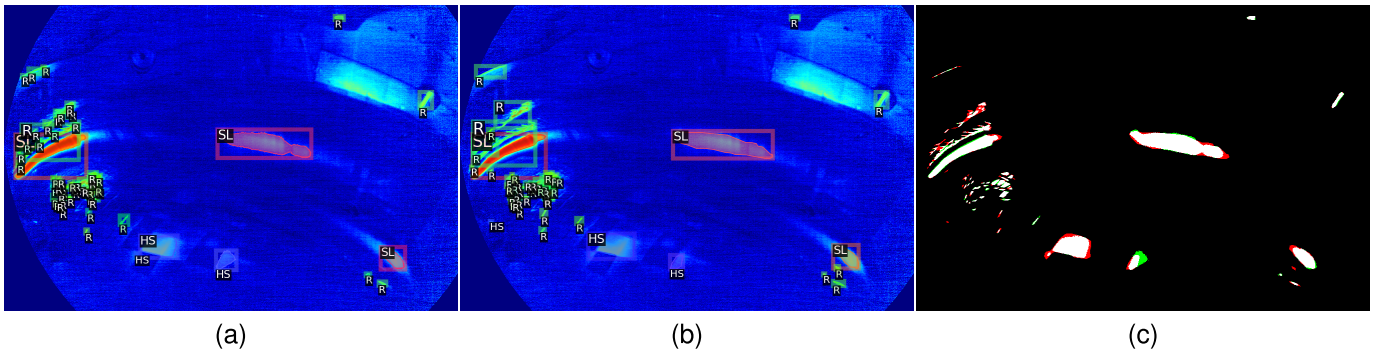


Fig. 5. 20180904.007 (AEF10), frame 810. (a) Generated annotations. (b) Ground truth annotations. (c) Difference (SDC: 0.847, tlwSDC: 0.883).

III. RESULTS

A. Quantitative

Quantitative results are presented in Table I. For comparison, similarity scores were also computed by replacing the proposed segmentation method with the global Otsu's thresholding [28] and local adaptive Gaussian thresholding [22]. Any mask is a score that merges all thermal events without considering their categories. SL, reflection, and hot-spot (HS) masks are scores that compare masks considering only one thermal event category, e.g., keep in generated and ground-truth masks only instances that are reflections. UFOs are not evaluated since only a single discharge sequence containing them has been identified.

B. Qualitative

1) W7-X: Qualitative results for a standard (EJM), high-iota (FTM), and low-iota (DBM) configuration are shown in Figs. 4–6, respectively. Green color marks over-segmentation and red color marks under-segmentation. For clarity, images are cropped to their reduced FoV, are contrast-enhanced,

ignored blobs are not visualized, and the thermal event category labels are abbreviated as follows: SL, reflection (R), and HS.

2) WEST: A qualitative result for a 16-bit 512×640 wide-angle view image from the WEST tokamak is shown in Fig. 7, which corresponds to bounding boxes in [11, Fig. 3]. Due to differences in established ontologies of thermal events, the radiated heat flux is substituted with UFO and electron type 1 with HS. Additionally, in W7-X, the SL substitutes inboard and outboard strike points, as the divertor region affected by a SL is determined from the scene model component map. No modifications were introduced in the method to apply it to this image.

IV. DISCUSSION

According to the quantitative results shown in Table I, the proposed semi-automatic annotation method, in conjunction with the proposed segmentation method, attains the highest average similarity to the manually annotated images and the lowest sample standard deviation. Low sample standard deviation indicates consistent performance between different

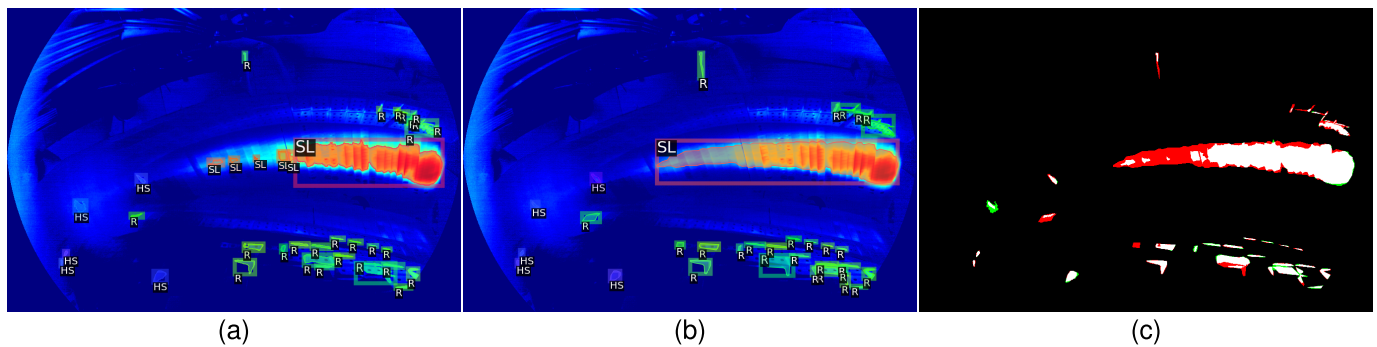


Fig. 6. 20181004.032 (AEF51), frame 1191. (a) Generated annotations. (b) Ground truth annotations. (c) Difference (SDC: 0.799, tlwSDC: 0.901).

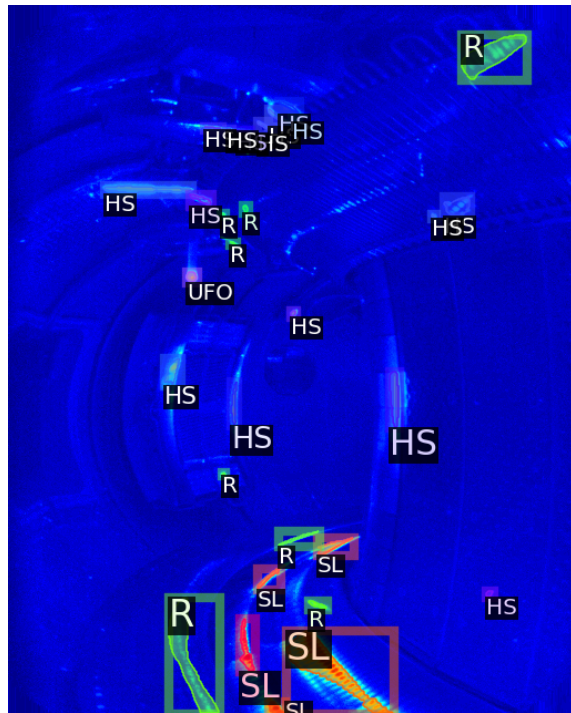


Fig. 7. WEST pulse 56927, DIR22, frame 466. Generated annotations. The ground-truth annotations are in [11, Fig. 3].

discharges. The highest segmentation performance is observed for SLs, which have the highest temperatures and relatively the largest areas. The lowest score is for reflections, which have the lowest temperatures, are the most numerous, and are scattered throughout FoV. The global Otsu's thresholding cannot segment thermal events of varying temperatures; therefore, it primarily segments the hottest SLs, which is reflected in its performance. The local adaptive Gaussian thresholding yields lower similarity compared to the proposed segmentation method, yet it is still relatively high. In some cases, it is advantageous as it considers a neighborhood of every pixel, not connected component levels. However, it is less consistent and prone to over-segmentation. In all cases, tlwSDC yielded higher similarity scores than SDC, which is expected as the segmentation methods prioritize higher temperature regions, which pose a greater risk to PFCs [2]. The smallest difference between SDC and tlwSDC is for reflections, as they are more uniform and often have no localized hot center radially spreading toward colder boundaries.

From qualitative results, it is visible that the annotations differ primarily on the edges of segmented instances. It is especially relevant since the contours of thermal events are fuzzy, and there is no perfect segmentation. As a consequence, border discrepancies are expected, rendering tlwSDC a more suitable metric. Although the method was developed for the W7-X stellarator, it did not require any modification to annotate the same thermal events with bounding boxes as in the WEST image in [11, Fig. 3] and a few more potentially relevant thermal events, simultaneously providing segmentation masks.

Annotation consistency is imposed by minimizing human involvement in the process and delegating only the classification stage. Considering a typical discharge sequence of 2000 images (20 s), the manual effort is reduced to assigning categories to a single image, and no manual segmentation is performed. The time to annotate an entire discharge sequence is reduced to hundreds of seconds, whereas fully manual annotation of a single ground-truth image takes half an hour on average. Moreover, a trained instance segmentation model on a smaller subset of generated annotations could assign initial categories, and the expert would only correct misassigned categories, further simplifying the process. The segmentation method could also be integrated with manual workflow to propose contours to the annotator for further manual adjustments.

A. Limitations

The following limitations are acknowledged. Temporal inconsistencies in contour delineation between successive frames result in sporadic contour flickering as no temporal information is exploited within the annotation method, particularly in colder thermal events whose boundaries are affected by noise. Further, the topology of surface temperature determines thermal event connectivity (except for leading edges that are processed separately). A significant surface temperature gap results in detached thermal events, while continuous surface temperature results in a grouped thermal event. It also limits the range of values within elongated events, i.e., colder tails of SLs are not segmented; otherwise, the local component cutoff level would have to be lowered to the level where the hotter part of the SL blends with the background hotter than the colder tail. However, colder parts are not essential for safety. Last but not least, to handle overlap in a reference image introduced by moving thermal events such

as UFOs, a discharge sequence can be split into pre-event and post-event to have separate reference images without overlap.

V. CONCLUSION

This study addresses the absence of annotations for complex infrared images from fusion devices. It is confirmed that the deterministic image-processing method based on a reference image and max-tree representation reduces annotation efforts and improves annotation consistency while maintaining high similarity to manual annotations. The proposed method achieved a mean SDC of 0.825 with a sample standard deviation of 0.030. Furthermore, the customized metric *tlwSDC*, which is better aligned with the objectives of machine protection and control, was proposed, and the method achieved a mean *tlwSDC* of 0.904 with a sample standard deviation of 0.018. Small score deviations show consistent performance between discharge sequences. The consistency of the annotations is preserved by minimizing the required human involvement and not requiring adaptation of method parameters. The segmentation stage is automatic, and the classification stage requires the expert only to classify thermal events in a single reference image to produce annotations for thousands of infrared images in a discharge sequence. The possible applicability to other fusion devices, e.g., the WEST tokamak, was indicated as well.

The proposed method enabled the generation of the first infrared dataset of annotated W7-X images. Facilitating, for the first time, the training of an instance segmentation model, i.e., mask region-based convolutional neural network (Mask R-CNN) [29], for this device. The next step is to research various resulting instance segmentation models for fusion devices.

ACKNOWLEDGMENT

This scientific article has been completed while Bartłomiej Jabłoński was a Doctoral Candidate in the Interdisciplinary Doctoral School at the Lodz University of Technology, Poland. Views and opinions expressed are however those of the author(s) only and do not necessarily reflect those of the European Union or the European Commission. Neither the European Union nor the European Commission can be held responsible for them. The Wendelstein 7-X Team is I. Abramovic, P. Agostinetti, M. Agredano Torres, S. Äkäslompolo, J. Alcuson Belloso, P. Aleynikov, K. Aleynikova, M. Alhashimi, A. Ali, N. Allen, A. Alonso, G. Anda, T. Andreeva, C. Angioni, A. Arkhipov, A. Arnold, W. Asad, E. Ascasibar, M.-H. Aumeunier, K. Avramidis, E. Aymerich, S.-G. Baek, J. Bähner, A. Baillod, M. Balden, M. Balden, J. Baldzuhn, S. Ballinger, M. Banduch, S. Bannmann, A. Banon Navarro, A. Bañón Navarro, T. Barbui, C. Beidler, C. Belafdil, A. Bencze, A. Benndorf, M. Beurskens, C. Biedermann, O. Biletskyi, B. Blackwell, M. Blatzheim, T. Bluhm, D. Böckenhoff, G. Bongiovi, M. Borchardt, D. Borodin, J. Boscary, H. Bosch, T. Bosmann, B. Böswirth, L. Böttger, A. Bottino, S. Bozhnikov, R. Brakel, C. Brandt, T. Bräuer, H. Braune, S. Brezinsek, K. Brunner, S. Buller, R. Burhenn, R. Bussiahn, B. Buttenschön, A. Buzás,

V. Bykov, I. Calvo, K. Camacho Mata, I. Caminal, B. Cannas, A. Cappa, A. Carls, F. Carovani, M. Carr, D. Carralero, B. Carvalho, J. Casas, D. Castano-Bardawil, F. Castejon, N. Chaudhary, I. Chelis, A. Chomiczewska, J.W. Coenen, M. Cole, F. Cordella, Y. Corre, K. Crombe, G. Cseh, B. Csillag, H. Damm, C. Day, M. de Baar, E. De la Cal, S. Degenkolbe, A. Demby, S. Denk, C. Dhard, A. Di Siena, A. Dinklage, T. Dittmar, M. Dreval, M. Drevlak, P. Drewelow, P. Drews, D. Dunai, E. Edlund, F. Effenberg, G. Ehrke, M. Endler, D.A. Ennis, F.J. Escoto, T. Estrada, E. Fable, N. Fahrenkamp, A. Fanni, J. Faustin, J. Fellingner, Y. Feng, W. Figacz, E. Flom, O. Ford, T. Fornal, H. Frerichs, S. Freundt, G. Fuchert, M. Fukuyama, F. Füllenbach, G. Gantenbein, Y. Gao, K. Garcia, J.M. García Regaña, I. García-Cortés, J. Gaspar, D.A. Gates, J. Geiger, B. Geiger, L. Giudicotti, A. González, A. Gorlaev, D. Gradic, M. Grahl, J.P. Graves, J. Green, E. Grelier, H. Greuner, S. Groß, H. Grote, M. Groth, M. Gruca, O. Grulke, M. Grün, J. Guerrero Arnaiz, S. Günter, V. Haak, M. Haas, P. Hacker, A. Hakola, A. Hallenbert, K. Hammond, X. Han, S.K. Hansen, J.H. Harris, H. Hartfuß, D. Hartmann, D. Hathiramani, R. Hatzky, J. Hawke, S. Hegedus, B. Hein, B. Heinemann, P. Helander, S. Henneberg, U. Hergenhausen, C. Hidalgo, F. Hindenlang, M. Hirsch, U. Höfel, K.P. Hollfeld, A. Holtz, D. Hopf, D. Höschen, M. Houry, J. Howard, X. Huang, M. Hubeny, S. Hudson, K. Ida, Y. Igitkhanov, V. Igoshine, S. Illy, C. Ionita-Schrittwieser, M. Isobe, M. Jabłczyńska, S. Jablonski, B. Jagielski, M. Jakubowski, A. Jansen van Vuuren, J. Jelonnek, F. Jenko, F. Jenko, T. Jensen, H. Jenzsch, P. Junghanns, J. Kaczmarczyk, J. Kallmeyer, U. Kamionka, M. Kandler, S. Kasilov, Y. Kazakov, D. Kennedy, A. Kharwandikar, M. Khokhlov, C. Kiefer, C. Killer, A. Kirschner, R. Kleiber, T. Klinger, S. Klose, J. Knauer, A. Knieps, F. Köchl, G. Kocsis, Ya.I. Kolesnichenko, A. Könies, R. König, J. Kontula, P. Kornejew, J. Koschinsky, M.M. Kozulia, A. Krämer-Flecken, R. Krampitz, M. Krause, N. Krawczyk, T. Kremeyer, L. Krier, D.M. Kriete, M. Krychowiak, I. Ksiazek, M. Kubkowska, M. Kuczynski, G. Kühner, A. Kumar, T. Kurki-Suonio, S. Kwak, M. Landreman, P.T. Lang, A. Langenberg, H.P. Laqua, H. Laqua, R. Laube, S. Lazerson, M. Lewerentz, C. Li, Y. Liang, Ch. Linsmeier, J. Lion, A. Litnovsky, S. Liu, J. Lobsien, J. Loizu, J. Lore, A. Lorenz, U. Losada, F. Louche, R. Lunsford, V. Lutsenko, M. Machielsen, F. Mackel, J. Maisano-Brown, O. Maj, D. Makowski, G. Manduchi, E. Maragkoudakis, O. Marchuk, S. Marsen, E. Martines, J. Martinez-Fernandez, M. Marushchenko, S. Masuzaki, D. Maurer, M. Mayer, K.J. McCarthy, O. McCormack, P. McNeely, H. Meister, B. Mendelevitch, S. Mendes, A. Merlo, A. Messian, A. Mielczarek, O. Mishchenko, B. Missal, R. Mitteau, V.E. Moiseenko, A. Mollen, V. Moncada, T. Mönnich, T. Morisaki, D. Moseev, G. Motojima, S. Mulas, M. Mulsow, M. Nagel, D. Naujoks, V. Naulin, T. Neelis, H. Neilson, R. Neu, O. Neubauer, U. Neuner, D. Nicolai, S.K. Nielsen, H. Niemann, T. Nishiza, T. Nishizawa, T. Nishizawa, C. Nührenberg, R. Ochoukov, J. Oelmann, G. Offermanns, K. Ogawa, S. Okamura, J. Ölmans, J. Ongena, J. Oosterbeek, M. Otte, N. Pablant,

- N. Panadero Alvarez, N. Panadero Alvarez, A. Pandey, E. Pasch, R. Pavlichenko, A. Pavone, E. Pawelec, G. Pechstein, T. S. Pedersen, G. Pelka, V. Perseo, B. Peterson, D. Pilopp, S. Pingel, F. Pisano, B. Plöckl, G. Plunk, P. Pölöskei, B. Pompe, A. Popov, M. Porkolab, J. Proll, M.J. Pueschel, M.-E. Puiatti, A. Puig Sitjes, F. Purps, K. Rahbarnia, M. Rasiński, J. Rasmussen, A. Reiman, F. Reimold, M. Reisner, D. Reiter, M. Richou, R. Riedl, J. Riemann, K. Riße, G. Roberg-Clark, V. Rohde, J. Romazanov, D. Rondeshagen, P. Rong, L. Rudischhauser, T. Rummel, K. Rummel, A. Runov, N. Rust, L. Ryc, P. Salembier, M. Salewski, E. Sanchez, S. Satake, G. Satheeswaran, J. Schacht, E. Scharff, F. Schauer, J. Schilling, G. Schlisio, K. Schmid, J. Schmitt, O. Schmitz, W. Schneider, M. Schneider, P. Schneider, R. Schrittwieser, T. Schröder, M. Schröder, R. Schroeder, B. Schweer, D. Schwörer, E. Scott, E. Scott, B. Shanahan, G. Sias, P. Sichte, M. Singer, P. Sinha, S. Sipliä, C. Slaby, M. Slecicka, H. Smith, J. Smoniewski, E. Sonnendrücker, M. Spolaore, A. Spring, R. Stadler, D. Stańczak, T. Stange, I. Stepanov, L. Stephey, J. Stober, U. Stroth, E. Strumberger, C. Suzuki, Y. Suzuki, J. Svensson, T. Szabolics, T. Szepesi, M. Szücs, F.L. Tabarés, N. Tamura, A. Tancetti, C. Tantos, J. Terry, H. Thienpondt, H. Thomsen, M. Thumm, J.M. Traverso, P. Traverso, J. Tretter, E. Trier, H. Trimino Mora, T. Tsujimura, Y. Turkin, A. Tykhyi, B. Unterberg, P. van Eeten, B.Ph. van Milligen, M. van Schoor, L. Vano, S. Varoutis, M. Vecsei, L. Vela, J.L. Velasco, M. Vervier, N. Vianello, H. Viebke, R. Vilbrandt, G. Vogel, N. Vogt, C. Volkhausen, A. von Stechow, F. Wagner, E. Wang, H. Wang, F. Warmer, T. Wauters, L. Wegener, T. Wegner, G. Weir, U. Wenzel, A. White, F. Wilde, F. Wilms, T. Windisch, M. Winkler, A. Winter, V. Winters, R. Wolf, A.M. Wright, G.A. Wurden, P. Xanthopoulos, S. Xu, H. Yamada, H. Yamaguchi, M. Yokoyama, M. Yoshinuma, Q. Yu, M. Zamanov, M. Zanini, M. Zarnstorff, D. Zhang, S. Zhou, J. Zhu, C. Zhu, M. Zilker, A. Zocco, H. Zohm, S. Zoletnik, and L. Zsuga.
- [1] M. Jakubowski et al., "Overview of the results from divertor experiments with attached and detached plasmas at Wendelstein 7-X and their implications for steady-state operation," *Nucl. Fusion*, vol. 61, no. 10, Oct. 2021, Art. no. 106003.
- [2] B. Jablonski et al., "Implementation and performance evaluation of the real-time algorithms for Wendelstein 7-X divertor protection system for OP2.1," *Fusion Eng. Des.*, vol. 190, May 2023, Art. no. 113524.
- [3] F. Pisano et al., "Learning control coil currents from heat-flux images using convolutional neural networks at Wendelstein 7-X," *Plasma Phys. Controlled Fusion*, vol. 63, no. 2, Feb. 2021, Art. no. 025009.
- [4] A. P. Sitjes et al., "Observation of thermal events on the plasma facing components of Wendelstein 7-X," *J. Instrum.*, vol. 14, no. 11, Nov. 2019, Art. no. C11002.
- [5] W. Gu, S. Bai, and L. Kong, "A review on 2D instance segmentation based on deep neural networks," *Image Vis. Comput.*, vol. 120, Apr. 2022, Art. no. 104401.
- [6] C. Sun, A. Shrivastava, S. Singh, and A. Gupta, "Revisiting unreasonable effectiveness of data in deep learning era," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 843–852.
- [7] M. Cordts et al., "The cityscapes dataset for semantic urban scene understanding," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 3213–3223.
- [8] S. Budd, E. C. Robinson, and B. Kainz, "A survey on active learning and human-in-the-loop deep learning for medical image analysis," *Med. Image Anal.*, vol. 71, Jul. 2021, Art. no. 102062.
- [9] R. Brunet et al., "Infrared measurement synthetic database for inverse thermography model based on deep learning," *Fusion Eng. Des.*, vol. 192, Jul. 2023, Art. no. 113598.
- [10] B. Cannas et al., "Towards an automatic filament detector with a faster R-CNN on MAST-U," *Fusion Eng. Des.*, vol. 146, pp. 374–377, Sep. 2019.
- [11] E. Grelier, R. Mitteau, and V. Moncada, "Deep learning-based process for the automatic detection, tracking, and classification of thermal events on the in-vessel components of fusion reactors," *Fusion Eng. Des.*, vol. 192, Jul. 2023, Art. no. 113636.
- [12] T. Cheng, X. Wang, S. Chen, Q. Zhang, and W. Liu, "BoxTeacher: Exploring high-quality pseudo labels for weakly supervised instance segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 3145–3154.
- [13] A. Berg, J. Johnander, F. D. de Geveigney, J. Ahlberg, and M. Felsberg, "Semi-automatic annotation of objects in visual-thermal video," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. Workshop (ICCVW)*, Oct. 2019, pp. 2242–2251.
- [14] T.-Y. Lin et al., "Microsoft COCO: Common objects in context," in *Computer Vision—ECCV*, D. Fleet, T. Pajdla, B. Schiele, and T. Tuytelaars, Eds. Cham, Switzerland: Springer, 2014, pp. 740–755.
- [15] F. Pisano et al., "Tools for image analysis and first wall protection at W7-X," *Fusion Sci. Technol.*, vol. 76, no. 8, pp. 933–941, Nov. 2020.
- [16] L. Vincent, "Morphological grayscale reconstruction in image analysis: Applications and efficient algorithms," *IEEE Trans. Image Process.*, vol. 2, no. 2, pp. 176–201, Apr. 1993.
- [17] Y. Chang, C. Jung, P. Ke, H. Song, and J. Hwang, "Automatic contrast-limited adaptive histogram equalization with dual gamma correction," *IEEE Access*, vol. 6, pp. 11782–11792, 2018.
- [18] C. Tomasi and R. Manduchi, "Bilateral filtering for gray and color images," in *Proc. 6th Int. Conf. Comput. Vis.*, Jan. 1998, pp. 839–846.
- [19] X. Bai, F. Zhou, and B. Xue, "Image enhancement using multi scale image features extracted by top-hat transform," *Opt. Laser Technol.*, vol. 44, no. 2, pp. 328–336, Mar. 2012.
- [20] P. Salembier, A. Oliveras, and L. Garrido, "Antiextensive connected operators for image and sequence processing," *IEEE Trans. Image Process.*, vol. 7, no. 4, pp. 555–570, Apr. 1998.
- [21] R. Clemente Bonjour, "Detection and classification of thermal events in the Wendelstein 7-X," MA thesis, Technical Univ. Catalonia (UPC), Barcelona School Telecommun. Eng., Signal Theory Commun. Dept., Barcelona, Spain, 2020. [Online]. Available: <https://upcommons.upc.edu/handle/2117/335770>
- [22] B. Jablonski et al., "Evaluation of NVIDIA Xavier NX platform for real-time image processing for plasma diagnostics," *Energies*, vol. 15, no. 6, p. 2088, Mar. 2022.
- [23] A. G. Silva and R. D. A. Lotufo, "Efficient computation of new extinction values from extended component tree," *Pattern Recognit. Lett.*, vol. 32, no. 1, pp. 79–90, Jan. 2011.
- [24] J. Matas, O. Chum, M. Urban, and T. Pajdla, "Robust wide-baseline stereo from maximally stable extremal regions," *Image Vis. Comput.*, vol. 22, no. 10, pp. 761–767, Sep. 2004.
- [25] R. Souza, L. Rittner, R. Machado, and R. Lotufo, "Maximal max-tree simplification," in *Proc. 22nd Int. Conf. Pattern Recognit.*, Aug. 2014, pp. 3132–3137.
- [26] X. Chen, Z. Zhao, Y. Zhang, M. Duan, D. Qi, and H. Zhao, "FocalClick: Towards practical interactive image segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 1290–1299.
- [27] L. R. Dice, "Measures of the amount of ecologic association between species," *Ecology*, vol. 26, no. 3, pp. 297–302, Jul. 1945.
- [28] N. Otsu, "A threshold selection method from gray-level histograms," *IEEE Trans. Syst. Man, Cybern.*, vol. SMC-9, no. 1, pp. 62–66, Jan. 1979.
- [29] K. He, G. Gkioxari, P. Dollár, and R. Girshick, "Mask R-CNN," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2980–2988.

REFERENCES